



Conditional Statements

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This Week In Python

- 1 Use comparisons and Boolean values
- 2 Write if, elif and else branches
- 3 Test the edges of a rule

An order is at the discount threshold. Should the rule approve it?

What would the program need to know?

A Condition Produces True Or False

A decision begins with a test.

1. Use == to compare values.
2. Use >= and <= for inclusive thresholds.
3. Write down the exact boundary.

What happens at 499, 500 and 501 TRY?

Approve Or Reject An Order

Indentation marks the branch that runs.

```
stock = 8
requested = 5
if requested <= stock:
    print("Order approved")
else:
    print("Insufficient stock")
```

Change requested to 8, then 9.

Several Outcomes Need A Clear Order

Python checks branches from top to bottom.

1. Handle invalid input first.
2. Put the most specific rule before a general one.
3. Let else cover the remaining valid case.

Example: `examples/03_elif_bands.py`

Combine Requirements

and means both conditions must be true.

```
amount = 1200
member = True
if amount >= 1000 and member:
    print("15% discount")
elif amount >= 1000:
    print("5% discount")
else:
    print("Standard price")
```

Predict the result when member becomes False.

Boolean Logic Changes The Policy

Check the rule in plain English first.

1. and: both requirements are met.
2. or: at least one requirement is met.
3. not: reverse a Boolean value.

Write a sentence for each operator.

A Safe input Boundary

Reject a value that the rule cannot use.

```
score = int(input("Score: "))  
if score < 0 or score > 100:  
    print("Invalid score")  
elif score >= 60:  
    print("Pass")  
else:  
    print("Try again")
```

Test -1, 0, 59, 60 and 101.

Find The Unreachable Branch

Branch order can hide a valid rule.

```
amount = 1200
if amount >= 500:
    print("5% discount")
elif amount >= 1000:
    print("10% discount")
```

Why can the second message never appear? Fix the order.

Live Coding: Order Approval

Turn a written rule into decisions.

1. Reject nonpositive quantities.
2. Reject a request above available stock.
3. Give members 10% off from 500 TRY.

Test just below, exactly at and above 500 TRY.

Project A: Retail Order Policy

Combine member discounts and delivery thresholds.

`projects/project_a_order_policy.py`

Check the total at every threshold.

Project B: Service Priority Classifier

Classify requests using explicit service desk rules.

`projects/project_b_service_priority.py`

Explain which rule produced each priority.

In-Class Lab: Order Approval Policy

Your Challenge

Assignment: `week03/lab-quiz/lab_assignment.md`

Submit in your repository: `week03/lab-quiz/`

Build it. Check a boundary. Explain one decision.

GitHub Checkpoint

Lab and quiz: week03/lab-quiz/

Keep weekly homework in week03/ outside that folder.

```
git add week03/lab-quiz/ README.md  
git commit -m "week03: complete lab quiz"  
git push
```

Show the lab, one test and the commit.

See You Next Week